

Number Size Distribution, Mass Concentration, and Particle Composition of PM₁, PM_{2.5}, and PM₁₀ in Bag Filling Areas of Carbon Black Production

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Number size characteristics and PM₁₀ mass concentrations of particles emitted during the packaging of various kinds of carbon blacks were measured continuously in the bag filling areas of three carbon black plants and concurrently at ambient comparison sites. PM₁₀, PM_{2.5}, and PM₁ dust fractions were also determined in the bag filling areas. The filter samples were then analyzed for elemental and organic carbon. Comparisons of the measured number size distributions and mass concentrations during bag filling activities with those measured parallel at the ambient site and with those determined during nonworking periods in the work area enabled the characterization of emitted particles. PM₁₀ mass concentrations were consistently elevated (up to a factor of 20 compared to ambient concentrations) during working periods in the bag filling area. Detailed analysis revealed that the carbon black particles released by bag filling activities had a size distribution starting at ≈ 400 nm aerodynamic diameter (d_{ae}) with modes around 1 μm d_{ae} and > 8 μm d_{ae}. Ultrafine particles (< 100 nm d_{ae}), detected in the bag filling areas, were most likely attributed to noncarbon black sources such as forklift and gas heater emissions.

Keywords carbon black, nanoparticle, particle number concentrations, ultrafine

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It is generally known that excess exposure to airborne particulate matter may cause adverse health effects in workers. Therefore, occupational exposure guidelines are recommended by organizations like the American Conference of Governmental Industrial Hygienists, the American Industrial Hygiene Association, and the German MAK Commission. Limit values by these organizations are based on mass concentrations. Recently, however, epidemiologic studies have shown higher associations of negative health effects from ultrafine particles (UFP; < 100 nm).⁽¹⁾ Ultrafine particles may contribute to health effects due to their high number concentration or surface area, high deposition efficiency in the

pulmonary region, and high propensity to penetrate the epithelium. Even though the mechanisms and particle characteristics causing negative health effects are heavily debated, particle characterization in ambient air and work areas enable detailed toxicologic studies by producing a more accurate database of particle exposures.

To our knowledge, no studies have been conducted that assess the UFP exposure represented by number size distributions in the carbon black industry. Therefore, this comprehensive study was initiated to characterize the potential fine (PM₁₀, PM_{2.5}, PM₁; mass concentration) and ultrafine particulate matter (< 100 nm d_{ae} number concentration) that workers may be exposed to in the carbon black industry. Within this study, particle number size distributions, particle mass concentrations, and chemical composition (carbon fractionation) were determined.

Descriptions of the Investigated Processes and Sites

Carbon black manufacturing normally consists of three process steps: reactor, pelletizing, and packaging. In the reactor step, chemical reactions take place in a furnace reactor to produce the primary carbon black particles, which are formed by gas-to-particle conversion during incomplete combustion of carbon-containing fuels. Primary particle sizes within the reactor can range from 1–500 nm, with the most common particles in the 10 nm to 100 nm range. These primary particles rapidly aggregate due to the high number concentrations in the closed reaction area and become the primary structural entity of carbon black. Aggregates are highly branched strings made up of roughly spherical nodules (primary particles) fused rigidly to one another. Aggregates may be made up of a few to many tens of nodules.

In the pelletizing step, very large agglomerates called “pellets” are formed by tumbling the newly made carbon black in a drum for a period of time or by spraying water on the carbon black in a pin mixer in such a way that the water facilitates

agglomeration. These agglomerates are then dried to make the final pellets.

Packaging, which is the phase of the manufacturing process investigated in this study, is the area where the product particles or pellets, are packed in small (≈ 25 kg) or large (≈ 1000 kg) bags for shipping.

METHODS

Measurement results discussed in this article were collected in sequence at two different locations in the packaging areas of each carbon black plant. The packaging equipment, ventilation, size of the packaging warehouse, and local wind patterns were quite diverse, significantly influencing the particle concentrations and ambient air mixing.

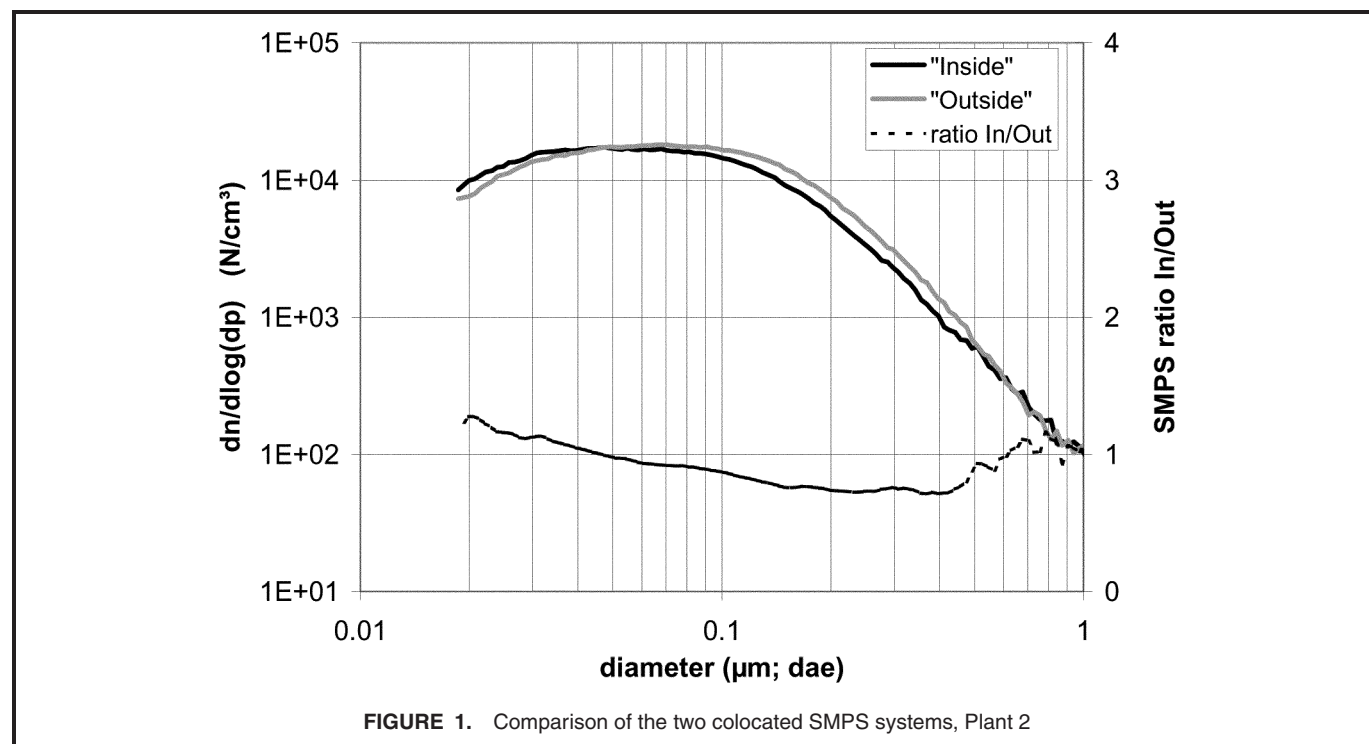
Particle size distributions were measured with a Scanning Mobility Particle Sizer ([SMPS] Platform 3080 series, Differential Mobility Analyzer 3081, Condensation Particle Counter 3025, and model 3077 Aerosol Neutralizer; TSI Inc., Shoreview, Minn.), which determines the particle number size distributions in the size range of 15–734 nm (Stokes diameter) by fractionating particles by their electrical mobility and subsequent counting and calculation of the concentration.⁽²⁾ No water vapor interference influenced these measurements since the conditions never exceeded the dew point of the conditions in the CPC 3025.

Ambient particle densities defined in the literature ranged from 1.5 to 1.9 g/cm³.⁽³⁾ Particle size distributions were converted to the aerodynamic diameter by assuming a particle density of 1.75 g/cm³. A particle size distribution was determined every 5 min. Two scanning mobility particle sizers (SMPS)

were employed at Plants 2 and 3 to simultaneously determine the ambient and work environment particle size distributions. As a result of an equipment malfunction, only one SMPS could be used at Plant 1. At this plant, the sampling protocol was modified, and consecutive sampling in work areas and at the ambient comparison site was conducted. The SMPSs were calibrated with latex particles of certified diameters prior to each set of measurements. The flow rate was checked regularly and adjusted if necessary. The two SMPS systems were compared with each other prior to each set of measurements and were found to be equivalent (Figure 1). Taking an estimated uncertainty into account due to changing sample locations, differences of a factor of two or more for one size channel were considered significant when data were compared.

An Aerodynamic Particle Sizer ([APS], model 3310, TSI Inc.) was employed in the work areas to identify and characterize airborne particles in the 0.5–15 μm (d_{ae} , aerodynamic diameter; size range used 0.7–12 μm)^(4,5) range by drawing the aerosol through a nozzle that accelerated the particles. The velocity of the particles, determined in the APS, depend on the aerodynamic diameter of the particle. The APS was calibrated with latex particles of certified aerodynamic diameters prior to each set of measurements. The flow rate was checked regularly and adjusted if necessary. Average particle size distributions were recorded every 5 min.

Two automatic instruments (TEOM[®] 1400ab; Rupprecht & Patashnik, Albany, N.Y.)^(6,7) were used as continuous PM₁₀ mass concentration monitors, one measuring in the work area while the other was placed in an ambient comparison site. The measurement principle of the TEOM is based on an oscillating tapered element, that is kept at a constant temperature of



50°C to reduce any influences of changes in ambient conditions on the mass determination. PM₁₀ mass concentrations were recorded every 5 min and averaged. The TEOM PM₁₀ data were recorded without any correction factor or correction function. PM₁₀ mass concentration ratios between the two samplers (inside, outside) ranged between 0.83 and 1.19 when co-located during validation. Considering the above comparison and an estimated uncertainty due to the frequent change of sampling locations for TEOM inside, mass concentrations were considered significantly different when inside and outside readings differed by a factor > 2.

Two LVS 3 (Derenda, Berlin, Germany) and one DHA 80 (DIGITEL, Zurich, Switzerland) filter sampling trains, equipped with PM₁₀, PM_{2.5}, and PM₁ inlets, respectively, were used in this study to determine the mass concentration and chemical composition of the different size fractions. The PM₁₀ and PM_{2.5} inlets met air sampling specifications described in U.S. and European ambient air quality standards.^(8–10) No standard concerning PM₁ exists to our knowledge. The volumetric flow rate of the filtration samplers was checked and inlets cleaned regularly. Preheated Munktell quartz fiber filters were used throughout the study to determine elemental carbon. Sampling durations ranged from 150 to 930 min.

Before and after sampling, all filters used in the study were equilibrated to 20 ± 1°C and 50 ± 5% relative humidity for at least 48 hours, then weighed 2 times within 12–24 hours between each weighing. If the weight difference was larger than 110 µg (equivalent 2 µg/m³ 24-hour sampling) for the filters sampled with the LVS 3 or larger than 1.44 mg (equivalent 2 µg/m³ 24-hour sampling) for filters sampled with the DHA 80, the filters were weighed a third time. If no agreement within these limits was achieved, filters were discarded.

Weighing was conducted in a climate-controlled cleanroom. Balances were calibrated before each weighing session and loaded filters were transported frozen in petri dishes or wrapped in aluminium foil. Filters were refrozen immediately after the final weighing to avoid losses during storage and prior to carbon analysis.

All instruments were inspected and calibrated in the laboratory prior to each set of measurements. Additionally, all instruments were co-located prior to and after the work area measurements at an ambient comparison site. This verified the comparability of the instruments. The terms “inside” and “outside” denote instrument use “in” the work areas or “outside” at an ambient comparison location. The ambient comparison sites chosen were on plant property with no obstacles within 50 m, had little local traffic near the site, and were at some distance (100–400 m) from the packaging operations.

Inside sampling locations were selected on the basis of proximity to appropriate packaging activities (1–3 m away) and on the basis of availability, accessibility, and safety. The samples represent areas where people work but do not represent personal breathing zone samples. All reported mass concentration data have been standardized to 0°C and 1.013 hPa.

Filter samples were analyzed for elemental and organic carbon to quantify the magnitude and proportion of the car-

bon fractions to the total particulate matter in the packaging areas. The purpose of this analysis was to determine whether identified ultrafine and particle mass concentrations could be attributed to carbon black. Elemental carbon (EC) and organic carbon (OC) were analyzed following preheating of the quartz fiber filters by a method developed by Cachier et al.⁽¹¹⁾ and Kuhlbusch.⁽¹²⁾

RESULTS AND DISCUSSION

Generally, three methods were used to determine specific sources in the bag filling areas. First, discontinuous or intermittent increases in particle concentrations could be identified by comparing the continuous measurement data before, during, and after a process activity. This comparison assumes that the air in the work area equilibrates relatively rapidly with ambient air conditions and was verified by comparing size distributions from the beginning to the end of break periods. Equilibration, mainly by ventilation, always occurred quickly. Secondly, permanent sources could be identified by comparing the ambient particle characteristics with those identified in the work area during longer nonworking periods. The final testing included carbon analysis, which helps determine whether carbon black is the possible source for any of the observed increases in PM_x mass concentrations.

Number Size Distributions

Plant 1

Particle number concentrations for three size classes (10–100 nm, 200–700 nm, and 1–10 µm), which corresponded to the three modes (nucleation mode < accumulation mode < coarse mode), were calculated from the number size distribution. These concentrations identified concentration changes with time in the three general modes, as shown in Figure 2a. Figure 2b gives the time series of the PM₁₀ mass concentrations both inside and outside, which clearly show the beginning and end of the work shift. Information on work processes, such as bag filling, shift breaks, etc., were recorded manually. The periods termed “event” (Figure 2a) show elevated coarse mode (mainly mechanically derived particles; >1 µm d_{ae}) particle number concentrations. Because these “events” were observed in all work areas and at the ambient location, they have to be attributed to a source outside of the plant.

The number concentrations of the >1 µm size class in Figure 2a clearly identify the start, break, and end points, whereas no significant changes are observed for the UFP. This type of graph was also used to identify the time periods for size distributions during work and nonwork periods.

Figure 3 shows a typical number size distribution during bag filling and work break periods. The particle number concentration ratios of work and work breaks (Figure 3) increase at about 400 nm d_{ae} with a first maximum at ≈ 1 µm followed by a further increase toward > 8 µm. This comparison analysis was conducted for several work break periods and for the start and

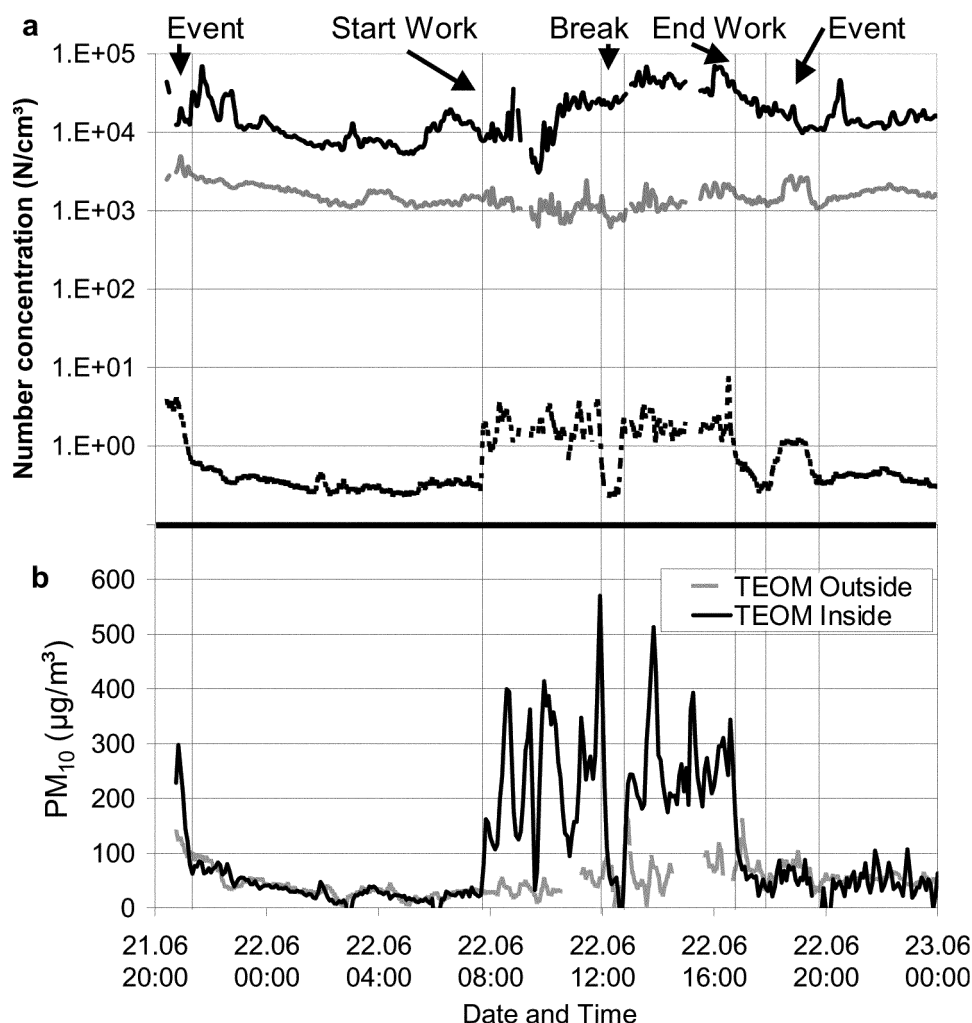


FIGURE 2. Time series of (a) number concentrations; top tracing: d_{ae} 0.015 to 0.10 μm ; middle tracing: d_{ae} 0.2 to 0.7 μm ; lower tracing: 2.5 to 10 μm ; and (b) PM_{10} mass concentrations over 36 hrs in Plant 1

end of work periods. All analyses showed similar results, with increases in particle number concentration ratios beginning at about 400 nm. This suggests the bag filling activity only emits particles > 400 nm.

Plant 2

Figure 4 shows typical number size distribution ratios of the closed packaging area during the filling of small bags (25 kg) compared with both an overnight nonwork period and concurrent outside size distributions. The work/night ratio begins to increase at about 400 nm d_{ae} and reaches a maximum at 1.2 μm and > 10 μm d_{ae} . This increase is most likely due to the bag filling operation. Also, a slight increase in the number concentration ratio below 50 nm can be observed, though no maximum was identifiable. The second ratio (size distribution work/ambient) shows a similar change in the size distribution as the work/night ratio indicating a source of particles < 100 nm. The increase of the number of particles > 400 nm is also evident in this ratio. The same ratio (work area/ambient)

for the nonwork period overnight (Figure 4) does not show these same differences in the size distribution. The ratios in this size range > 400 nm are similar to the ones shown in Plant 1, which indicate a strong source of coarse mode particles due to bag filling activities, though additional sources of UFP are seen in Plant 2.

Figure 5a shows the particle number concentrations, and Figure 5b shows the number size distribution during the filling of large bags (super sacks, 1000 kg). The particle counts determined with the APS (0.7–14 μm d_{ae}) are significantly lower than those determined with the SMPS in the overlapping size range (0.7–0.9 μm) (Figure 5b). The two instruments were placed about 2 m apart with the SMPS about 1.5 m away from the packing machine on the sampling platform, while the APS was 3.5 m away on the ground, sampling at 1 m lower in height. These different sampling locations, in relation to the packing machine, caused differences in the overlapping particle diameter range as a result of dilution effects on the particles coming from this source.

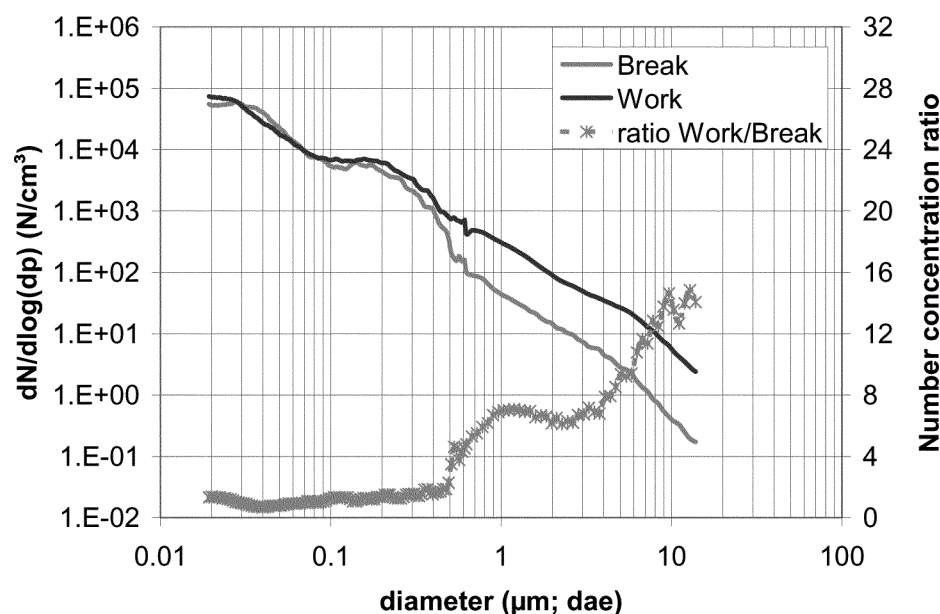


FIGURE 3. Number size distributions during bag filling and breaks in Plant 1

Regular peaks in the number concentrations below 100 nm were observed to coincide with large bag filling (Figure 5a). A detailed analysis of the bag filling process and comparison with the manual records of activities enabled us to differentiate the time periods associated with bag filling and bag change-out. The distributions (bag change, filling) and ratios (Figure 5b) are representative measurements of 22 bag change and filling events.

The rapid changes of the ultrafine particle concentrations shown in Figure 5a and indoor ultrafine and accumulation mode particle number concentrations are similar and show that (1) ultrafine particles do not stay airborne in the bag filling area for prolonged periods of time, (2) very few accumulation mode particles are emitted by the bag filling activities, and (3) ultrafine particles are emitted only during bag changing activities. Forklifts were active during the changing of large

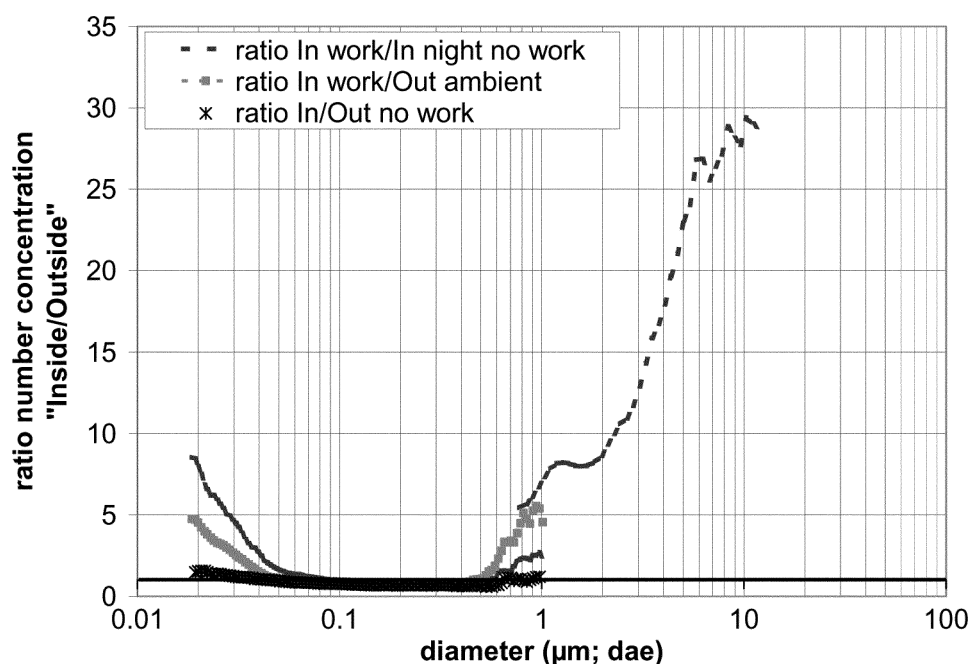


FIGURE 4. Number size distribution ratios for bag filling E 570 (small bags) and nonworking period night time in Plant 2

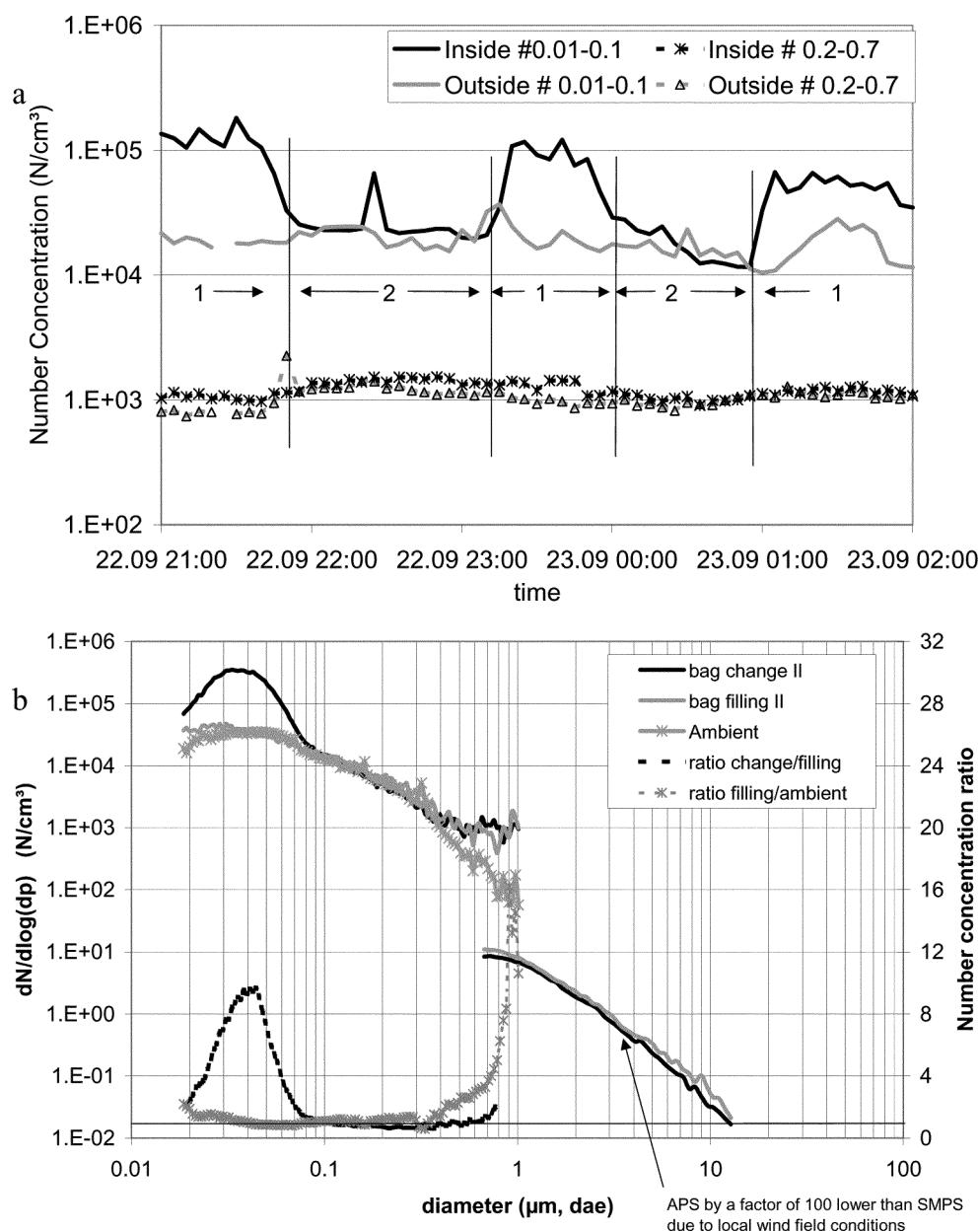


FIGURE 5. (a) Time series of particle number concentrations showing changes for ultrafine particles (10–100 nm) and accumulation mode particles (200–700 nm) during filling large bags in Plant 2; 1 denotes bag change and 2 denotes bag filling. (b) Corresponding average number size distribution

bags but were not used during bag filling operations. Hence, measurements were made with only a forklift running with no ongoing bag filling activities.

Figure 6 shows the number size distributions during a work break and the operation of a single forklift (propane fuel) near the sampling instrumentation. There were no other activity changes between the break time and the running of the forklift. The size distribution comparison in Figure 6 clearly shows an increase in the UFP size range 20–50 nm d_{ae} as well as in the larger particle size range ($>1 \mu\text{m}$). Both emissions may be attributed to the forklift activity since there were no other

ongoing working activities. The particles in the UFP size range are emitted with the exhaust of the forklift engine whereas the larger particles are most likely resuspended particles due to air turbulence from the forklift motion.

Plant 3

Figure 7 shows number size distribution ratios for work/no work and no work/ambient in the bag filling area. Two different time periods were chosen as an example. The first was during bag filling of small bags and the second during a work break just prior to the bag filling.

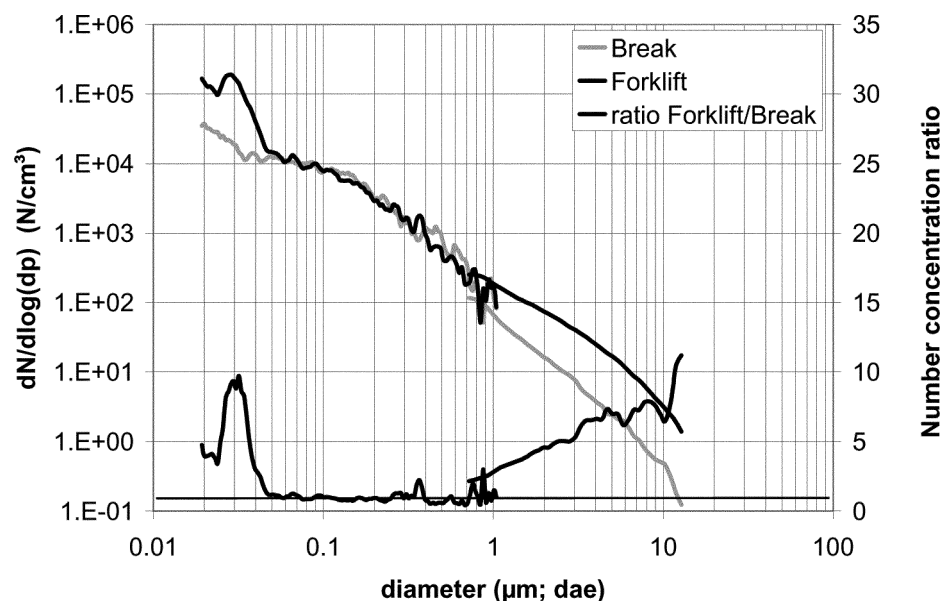


FIGURE 6. Number size distribution during forklift use in Plant 2

Figure 8 shows number size distributions obtained during filling of large bags and a work break prior to bag filling. The ratio shown in this case is the one of bag filling/break time.

Figures 7 and 8 show two modes during bag filling. The first is around 30–50 nm and the second is for particles > 400 nm and > 1 μm, respectively. The latter mode correlates with the ones identified in Plant 1 and Plant 2 and can be attributed to particle emissions due to bag filling activities.

The mode (maximum) around 30–50 nm is well pronounced with ratios of about 20 and 60, respectively (Figures 7 and 8).

The increase of these particles in the UFP size range was consistently identified only during bag filling activities. A source of particles in this mode was investigated by recording forklift activities during bag filling that operated in close proximity to the measurement platform. The forklifts used in Plant 3 were powered by diesel engines. Figure 9 shows number size distributions during forklift operation (no bag filling activity) near the instrumentation and a subsequent break period. The ambient size distribution shown is the average during the forklift operations, with the work break occurring 2 hours after the

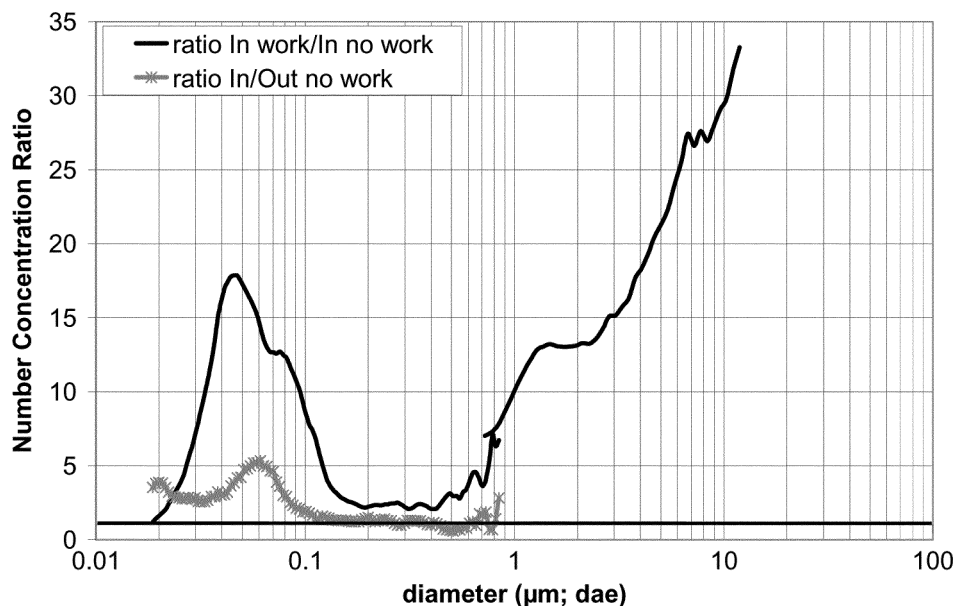
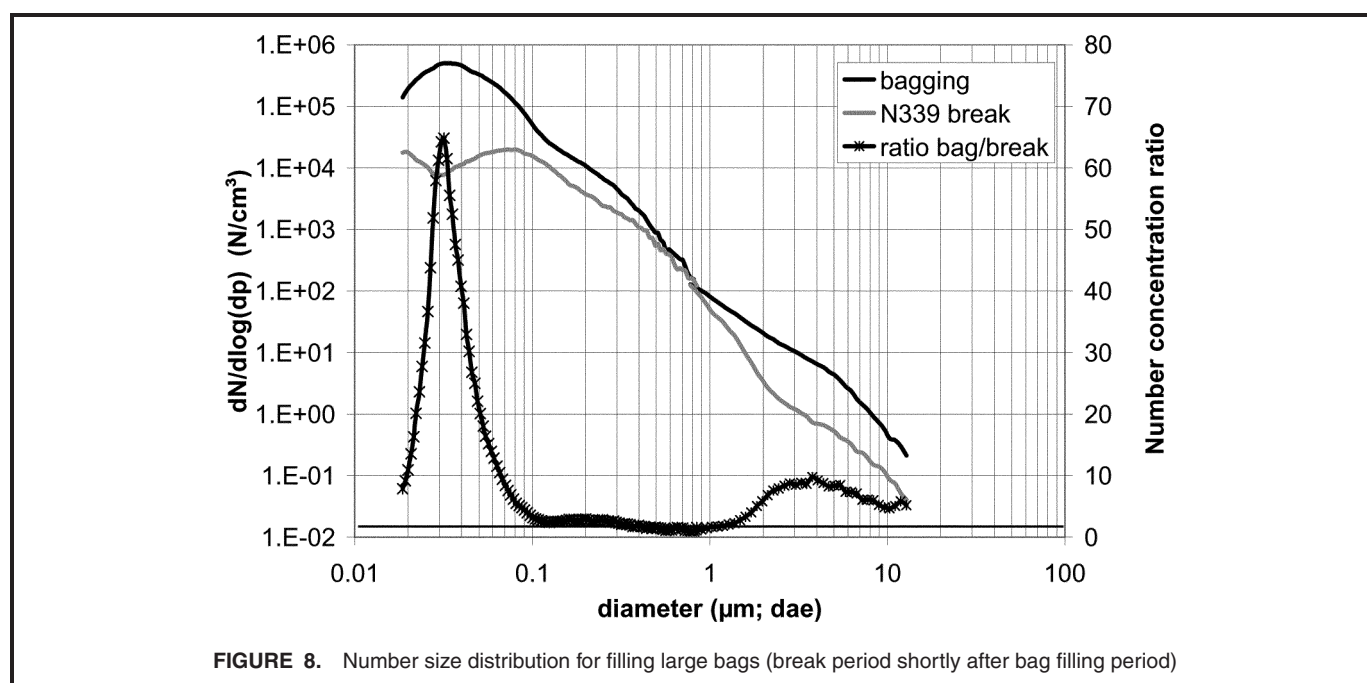


FIGURE 7. Number size distribution ratios for filling small bags, Plant 3

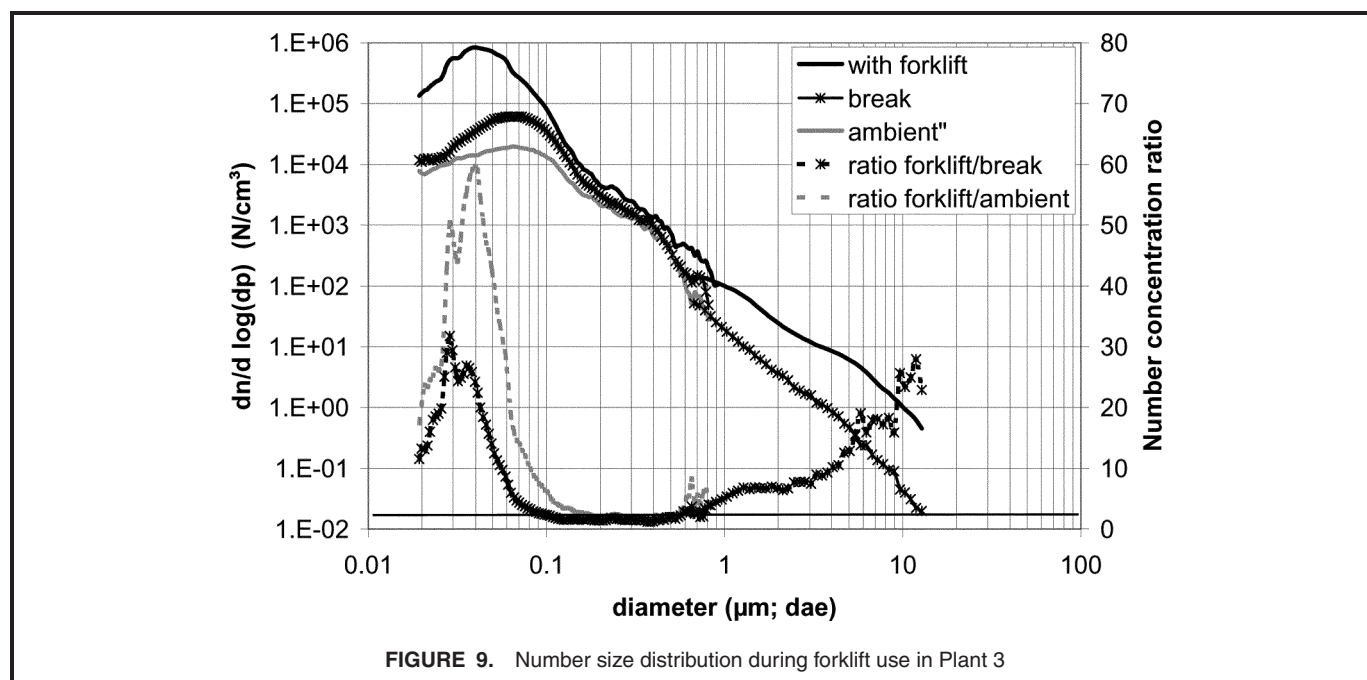


forklift operations. The comparison ratios shown in the same figure demonstrate that the forklift emissions are the source of the 30–50 nm particles identified during bag filling activities.

The differences in the particle size distributions in the ultra-fine particle size range in Figure 7, compared with Figures 8 and 9, appear to be due to different forklift activities during the sampling periods. Several forklifts were active at various distances to the sampling area in the packaging warehouse during the small bag filling activities (Figure 7). Conversely, only one forklift was used regularly in the vicinity of the sampling site during filling of the large bags (Figure 8) and

during the isolated testing of the one fork lift (Figure 9). It has been frequently observed that particle size distributions will have several modes if diesel emissions come from different engines because of different airborne suspension times.⁽¹³⁾

Figure 10 shows an example (6-hour period) of the number size distributions measured inside and outside during longer periods at night when no activities were taking place in the packaging warehouse. The number concentration ratio shown in this figure indicates a source of particles with a mode < 30 nm and a concentration ratio of > 10. This can also be seen in Figure 7 for the ratio break In/Out and was consistently



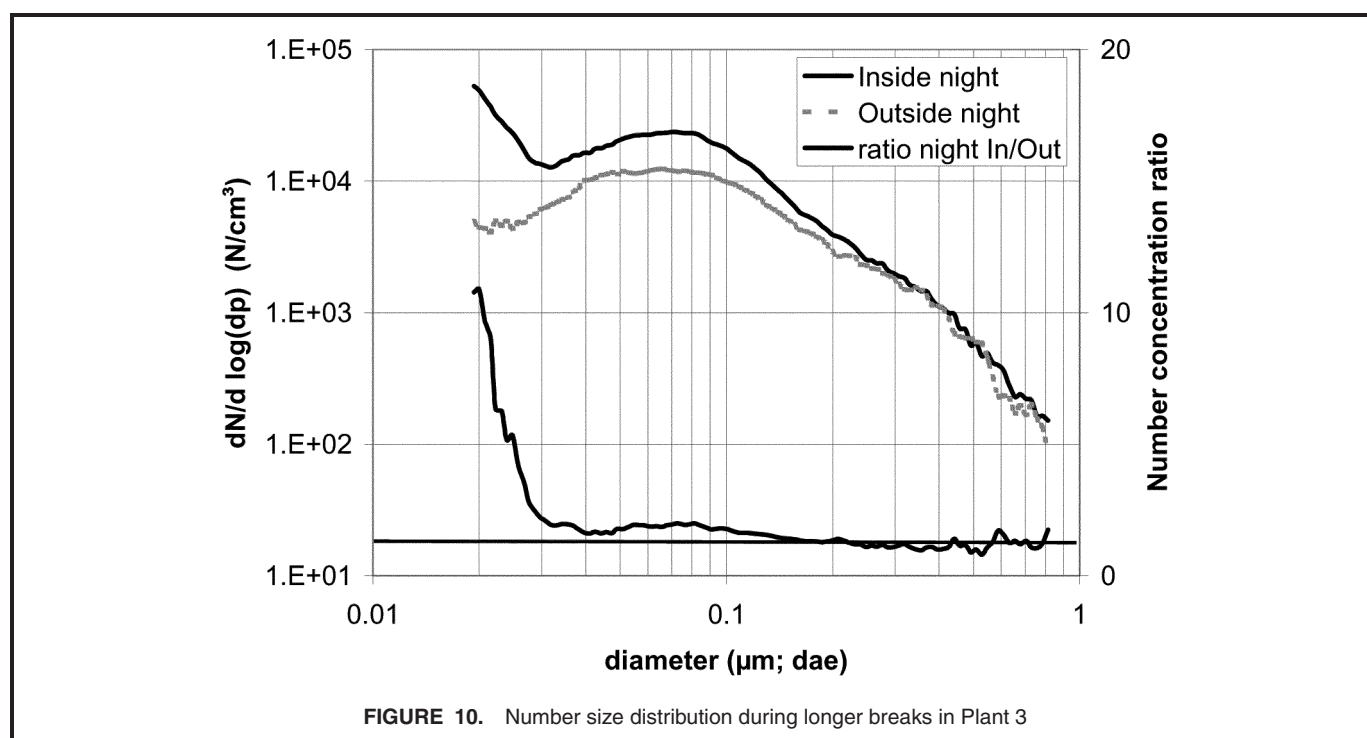


FIGURE 10. Number size distribution during longer breaks in Plant 3

observed during longer periods without any activities. Since this increase could be observed consistently only during longer break periods, a permanent process is likely to be the source of these UFPs. The source is expected to be close to the measurement equipment since the ratio increases down to the lower detection limit (17 nm d_{ae}) of the instrument. The most likely source for these particles is the butane space heaters used for comfort heating of the packaging warehouse. One of these burners, in fact, was placed about 5–10 m away from the measurement platform. No gas heaters were used in either of the other plants.

Summary Particle Number Size Distributions

In each plant, the number size distribution comparisons identified particles $> 0.4 \mu\text{m } d_{ae}$. Particle emissions in the ultrafine range ($< 100 \text{ nm}$) were observed in Plant 2 and 3 but not in Plant 1. Electric forklifts were used in Plant 1, propane forklifts in Plant 2, and diesel forklifts in Plant 3. Comparison of crude forklift emission measurements with size distributions identified in the bag filling areas correlated well, implicating the forklifts as the source of increased UFP. Additionally, butane space heaters were the likely continuous source of ultrafine particles in Plant 3. Hence, carbon black particles emitted during bag filling have size characteristics with particle sizes starting at $\approx 0.4 \mu\text{m}$ in aerodynamic diameter with modes at particle sizes of $1\text{--}2 \mu\text{m } d_{ae}$ and $> 8 \mu\text{m } d_{ae}$.

MASS CONCENTRATIONS

Table I provides an overview of the mass concentrations identified at the ambient site and the different bag filling

areas using the TEOM and filter samplers at each of the three plants.

Work area/ambient ratios of PM_{10} mass concentrations are provided in Table II. Comparing the mass concentration ratios for nonwork evening shift to the ambient concentration (103% and 119%, no work under way) determined that no significant increase in PM_{10} mass concentration indoors were present. Table II gives the $\text{PM}_{2.5}/\text{PM}_{10}$ and $\text{PM}_1/\text{PM}_{10}$ mass concentration ratios.

The continuous mass concentration readings clearly showed the differences between bag packing and work break times, which indicated that the mass concentration changes were due to the bag packing process (Figure 2b). The $\text{PM}_1/\text{PM}_{10}$ and the $\text{PM}_{2.5}/\text{PM}_{10}$ mass concentration ratios showed the lowest value for the bag filling area during times with higher mass concentrations (Table II). This indicates the higher mass concentrations in the bag filling area are due mainly to coarse mode particles ($> 1 \mu\text{m } d_{ae}$) that correspond to the measurements of the number size distributions.

CARBON FRACTIONATION

Table III provides a summary of the PM_X and organic, elemental, and total carbon concentrations from each of the three bag filling areas. The PM_{10} noncarbon fraction was calculated by subtracting the determined total carbon (TC) mass concentration from the corresponding PM_X particle mass concentration. Comparing noncarbon fraction values in the bag filling areas with those at the corresponding comparison site show that they are quite similar. This indicates that carbon is the main source of the additional mass concentrations.

TABLE I. Mass Concentrations in Plants 1, 2, and 3

	Start Time	End Time	TEOM PM ₁₀ μg/m ³	PM ₁₀ μg/m ³	PM _{2.5} μg/m ³	PM ₁ μg/m ³	PM _{2.5} / PM ₁₀	PM ₁ / PM ₁₀
Plant 1								
Ambient air	27.06.00 14:05	27.06.00 22:35	45 ^A	34	20	— ^B	0.59	—
Ambient air	27.06.00 23:22	28.06.00 08:58	32	40	23	— ^B	0.58	—
Ambient air	28.06.00 11:33	28.06.00 18:00	67 ^A	62	31	— ^B	0.50	—
Bag filling area, no bagging	21.06.00 21:00	22.06.00 07:11	40	49	22	24	0.46	0.50
Bag filling area N772 ^C	20.06.00 08:55	20.06.00 14:55	— ^B	118	81	47	0.69	0.40
Bag filling area N234	21.06.00 06:07	21.06.00 10:15	216	203	74	40	0.36	0.20
Bag filling area N234	21.06.00 10:35	21.06.00 16:48	370	231	70	41	0.30	0.18
Bag filling area N550	22.06.00 09:17	22.06.00 15:12	256	340	113	35	0.33	0.10
Plant 2								
Ambient	19.09.00 10:10	19.09.00 17:07	21	38	24	16	0.63	0.43
Ambient	19.09.00 17:23	20.09.00 05:22	10	13	12	13	0.88	0.97
Bag filling, no bagging	20.09.00 18:02	21.09.00 05:50	19	27	12	14	0.45	0.53
Bag filling E 570, BP 460	20.09.00 06:18	20.09.00 15:58	84	73	56	33	0.77	0.45
Bag filling E 570, E 410	21.09.00 06:28	21.09.00 12:42	99	94	55	47	0.58	0.50
Bag filling fluffy	21.09.00 17:53	22.09.00 09:21	55	85	53	20	0.63	0.24
Bag filling fluffy	22.09.00 09:38	22.09.00 15:42	126	137	71	37	0.52	0.27
Bag filling fluffy	23.09.00 05:40	23.09.00 08:16	38	40	23	18	0.58	0.46
Plant 3								
Ambient	14.11.00 14:46	14.11.00 20:26	10	4	3	2	0.75	0.57
Ambient	22.11.00 16:23	23.11.00 08:42	^D	15	14	12	0.95	0.77
Bag filling N550 (small bags)	15.11.00 13:32	15.11.00 21:29	130	63	31	46 ^E	0.49	0.72
Bag filling N339 (small bags)	15.11.00 21:44	16.11.00 08:53	71	76	64	52	0.84	0.68
Bag filling N339, HV3396 (large bags)	16.11.00 12:59	16.11.00 22:24	165	53	28	280 ^E	0.52	5.26
Bag filling HV 3396 (large bags)	16.11.00 22:39	17.11.00 08:40	45	44	14	181 ^E	0.33	4.16

Note: Concentrations standardized to 0°C and 1.013 hPa.

^ANumbers in bold indicate values influenced by power failure, ca. 2 hours.

^BMalfunction of instrument.

^CDenotes ASTM designation for manufactured carbon black.⁽¹⁴⁾

^DReadings negative.

^EPM₁ elevated due to location of the sampler too close to the source, PM₁ about 80 cm from the source, PM_{2.5} and PM₁₀ between 2–3 m from the source.

TABLE II. Average Mass Concentration Ratio Working Place to Outside and Average Mass Size Ratios at Plants 1, 2, and 3

	TEOM In/TEOM Out	PM _{2.5} /PM ₁₀	PM ₁ /PM ₁₀	No. Obs.
Plant 1				
Ambient (co-located)	0.97	0.56	^A	3
Bag filling area 2 (no work)	1.03	0.46	0.50	1
Bag filling area 1		0.69	0.40	1
Bag filling area 2	6.20 (max 20.00)	0.33	0.16	3
Plant 2				
Ambient (co-located)	1.19	0.75	0.70	2
Bag filling area 1 (no work)	1.19	0.45	0.52	2
Bag filling area 1	8.09 (max 11.00)	0.68	0.49	1
Bag filling area 2	4.18 (max 12.00)	0.58	0.32	3
Plant 3				
Ambient (co-located)	0.94	0.85	0.67	2
Bag filling area 1	7.97	0.66	0.71 ^B	2
Bag filling area 2	6.62	0.43	4.71 ^B	2

Note: Max. = maximum ratio of 30 min values to manual filter time intervals.

^ASampler malfunction.

^BToo close to source.

TABLE III. PM_x Carbon Fractionation in Plants 1–3

	PM ₁₀ μg/m ³	PM ₁₀ no carbon μg/m ³	TC/PM ₁₀ (%)	EC/PM ₁₀ (%)	PM _{2.5} μg/m ³	TC/PM _{2.5} (%)	EC/PM _{2.5} (%)	PM ₁ μg/m ³	TC/PM ₁ (%)	EC/PM ₁ (%)
Plant 1										
Ambient	45	28	38 ± 2	12	25	57 ± 4	14	Sampler defect		
Bag filling area 2 (no work)	49	26	47	20	22	74	35	24	62	13
Bag filling area 1	118	43	64	46	81	79	40	47	68	30
Bag filling area 2	258	20	92 ± 5	79	85	99 ± 2	75	39	92 ± 5	52
Plant 2										
Ambient	25	22	15	9	18	34	21	14	22	9
Bag filling area 1 (no work)	27	13	51	47	12	77	73	14	77	47
Bag filling area 1	83	19	76	74	55	79	94	40	88	81
Bag filling area 2	75	22	67	53	42	82	60	24	75	51
Plant 3										
Ambient ^A	15	11	30	16	14	28	21	12	23	11
Bag filling area 1	70	–4	107	92	48	99	104	49	96	71
Bag filling area 2	48	4	90	71	28	106	101	231 ^B	71 ^B	64 ^B

Note: ± = absolute standard deviation.

^AOnly one filter series with concentrations above detection limit.

^BPM₁ sampler much closer to the source than the other two.

Increments in PM and carbon mass were calculated by subtracting the ambient concentrations from the concentrations measured in the bag filling area as an estimate of the contribution of elemental carbon to the additional mass. From this calculation we found that elemental carbon accounts for (90 ± 9)%, (92 ± 25)%, and (81 ± 12)% of the additional PM₁₀, PM_{2.5}, and PM₁ mass, respectively. Hence, the product particles are the likely source of the elevated mass concentrations in the bag filling areas. These results correspond well with those obtained from the number size distribution and PM mass concentration measurements during the bag filling activities.

SUMMARY AND CONCLUSIONS

Measurements in the bag filling areas of three plants enabled the characterization of the number size distributions of particles released by bag filling activities. Detailed analyses of size distributions in the bag filling area compared with those measured concurrently at the ambient site, and comparison of size distributions during work and break periods were performed. Both gave comparable results and showed two particle modes. During bag filling, the particle number concentrations increase for particles > 400 nm with modes around 1 μm and > 8 μm. Ultrafine particle emissions (<100 nm d_{ae}) detected in the bag filling areas could be attributed to forklifts running either on propane or diesel. Another source of ultrafine particles were butane gas heaters in Plant 3.

Number size distribution and mass concentration changes could clearly be differentiated between work and break periods. Comparisons of the ambient samples to the bag filling

area samples PM₁₀ mass concentrations showed increases by factors of 4–8.

PM₁₀, PM_{2.5}, and PM₁ were also measured with size-selective filter samplers in the bag filling areas and at the ambient comparison site. When the PM₁/PM₁₀ and PM_{2.5}/PM₁₀ mass concentration ratios for periods with elevated mass concentrations were compared with the ratios at the ambient site, it was concluded that most of the additional mass concentration measured in the bag filling area was due to particles > 1 μm d_{ae}. This result correlates well with the number size distributions of particles being released during bag filling activities.

The filters used for the PM_x-mass concentration measurements were also analyzed for organic and elemental carbon. Manufactured carbon black was identified as the probable source of the elevated mass concentrations in the bag filling areas.

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